

Love Swarm Nullification Enhanced: A Leader-Guided Metaheuristic with Emotional Dynamics

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Abstract—Nature-inspired metaheuristics often mimic collective behaviors such as flocking or schooling, yet rarely incorporate emotional or relational dynamics. This paper presents the Love Swarm Nullification Enhanced (LSNE) algorithm — a novel population-based optimizer that simulates emotional bonds (love) and destructive negation (nullification) among agents, extended with a distinct leader agent. The leader guides the swarm via global attraction, balancing exploration and exploitation through adaptive love and nullification intensities. LSNE is evaluated on standard benchmark functions and compared with recent swarm algorithms. Results show that the leader mechanism significantly accelerates convergence while maintaining diversity, outperforming baseline Love Swarm variants and several state-of-the-art methods. The algorithm’s social dynamics and parameter sensitivity are analyzed, demonstrating robust performance across unimodal and multimodal landscapes.

Index Terms—swarm intelligence, metaheuristic, leader-follower, emotional algorithm, optimization

I. INTRODUCTION

Swarm intelligence (SI) algorithms have proven effective in solving complex optimization problems by simulating decentralized collective behavior observed in nature, such as ant colonies, bird flocks, and fish schools. Classic examples include Particle Swarm Optimization (PSO) [1], Ant Colony Optimization (ACO) [2], and Artificial Bee Colony (ABC) [3]. These methods rely on simple rules that govern individual agents, leading to emergent intelligent global behavior.

Recently, there has been growing interest in imbuing SI algorithms with emotional or social concepts. The Love Swarm algorithm [4] introduced the metaphor of “love” (attraction) and “nullification” (repulsion/elimination) among agents, where each agent is attracted to others with high fitness (love) and tends to nullify (remove or weaken) those with low fitness. This emotional dynamics enhances diversity and prevents premature convergence. However, the original Love Swarm lacks a centralized guiding force; all interactions are local, which can slow convergence in high-dimensional or rugged landscapes.

To address this limitation, we propose the **Love Swarm Nullification Enhanced (LSNE)** algorithm, which incorporates a **leader** agent. The leader acts as a global best-so-far point that exerts an attractive force (love) on all other agents, guiding the swarm towards promising regions. Additionally, the leader is immune to nullification and adaptively updates its intensity to balance exploration and exploitation. The leader mechanism transforms the purely distributed system into a

leader-follower architecture, where the leader provides directional information while the emotional interactions maintain swarm diversity.

The main contributions of this paper are:

- The design of the LSNE algorithm with a dedicated leader agent that interacts via enhanced love and nullification rules.
- Adaptive mechanisms for the leader’s influence factor and the swarm’s emotional intensities, controlled by population diversity.
- Comprehensive experimental validation on 23 benchmark functions, including comparison with original Love Swarm, PSO, GWO, and other state-of-the-art algorithms.
- Sensitivity analysis and ablation study on the leader’s role, demonstrating its impact on convergence speed and final accuracy.

The rest of the paper is structured as follows. Section II reviews related work in emotional swarm algorithms and leader-guided strategies. Section III details the LSNE algorithm, including the leader mechanism. Section IV presents experimental setup and results. Section V concludes with future directions.

II. RELATED WORK

A. Love Swarm and Emotional Metaheuristics

The Love Swarm algorithm [4] introduced the concepts of *love* and *nullification* into a swarm optimizer. In each iteration, agents compute a love score toward every other agent based on fitness difference and distance. An agent moves toward those it loves, and those that receive insufficient love are “nullified” (deleted or reinitialized). This mimics social relationships where entities bond with successful neighbors and abandon weak ones. Variants such as Love Swarm Nullification (LSN) [5] improved nullification rules to prevent diversity loss. Despite its novelty, the purely decentralized love interaction can result in slow convergence because agents lack a unified goal; they may wander in local clusters.

B. Leader-Based Swarm Intelligence

Introducing a leader or global guide is a well-known technique to accelerate convergence. In PSO, the global best particle serves as a leader towards which all particles are attracted, along with their personal best [1]. The Grey Wolf Optimizer (GWO) mimics a leadership hierarchy with alpha, beta, and delta wolves that guide the pack [6]. The Salp Swarm

Algorithm (SSA) uses a leader salp that moves toward the food source (best solution), and followers adjust based on chain dynamics [7]. These leader-follower structures often improve exploitation but may reduce diversity if the leader gets stuck in a local optimum. Hybrid approaches have combined global leaders with local interaction rules to mitigate this issue [8].

C. Gap and Our Approach

No existing work has integrated a leader into an emotion-based swarm like Love Swarm. We hypothesize that a carefully designed leader, coupled with adaptive love/nullification dynamics, can synergize the benefits of centralized guidance and emotional diversity maintenance. LSNE fills this gap by defining a leader agent that continuously influences the swarm while evolving through its own love–nullification interactions with the population.

III. METHODOLOGY: LOVE SWARM NULLIFICATION ENHANCED

A. Standard Love Swarm Nullification (Baseline)

We first recall the baseline Love Swarm Nullification (LSN) algorithm [5]. The swarm consists of N agents, each with a position $\mathbf{x}_i \in \mathbb{R}^D$ and a fitness $f(\mathbf{x}_i)$. At each iteration t , for every agent i , a *love matrix* L_{ij} is computed:

$$L_{ij}(t) = \begin{cases} \alpha \cdot e^{-\beta d_{ij}} \cdot \frac{f(\mathbf{x}_j) - f(\mathbf{x}_i)}{|f_{max} - f_{min}|}, & \text{if } f(\mathbf{x}_j) > f(\mathbf{x}_i) \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where $d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$, α and β are parameters controlling intensity and distance decay, and f_{max}, f_{min} are the current population's max and min fitness. An agent i loves j if $L_{ij} > \theta$ (a threshold). The agent then moves according to a weighted sum of love vectors:

$$\mathbf{v}_i^{\text{love}} = \sum_{j \in \mathcal{L}_i} L_{ij}(\mathbf{x}_j - \mathbf{x}_i) \quad (2)$$

Nullification: agents with total received love $R_j = \sum_i L_{ij}$ below a threshold ϕ are considered unloved and are nullified—either removed and replaced with a random new agent or reinitialized near loved agents. After movement, positions are updated: $\mathbf{x}_i(t+1) = \mathbf{x}_i(t) + \mathbf{v}_i^{\text{love}} + \epsilon \mathbf{r}$, where ϵ is a small random step for exploration.

B. Introduction of the Leader Agent

In LSNE, we designate one special agent as the **leader** ($\mathbf{x}_\ell$). The leader is part of the population but plays a dual role: it acts as a global attractor and also participates in love interactions. The leader's position is updated differently, combining attraction to the best solution ever found and influence from loved agents.

1) *Leader Selection and Update*: Initially, the leader is the agent with the best fitness. In subsequent iterations, if a non-leader agent achieves fitness better than the leader, it becomes the new leader. However, to avoid frequent switching, we introduce a tenure parameter T_ℓ , forcing the leader to remain for at least T_ℓ iterations before replacement, unless a significantly better solution is found (improvement $> \delta$). The leader's own movement follows:

$$\mathbf{x}_\ell(t+1) = \mathbf{x}_\ell(t) + \mathbf{v}_\ell^{\text{self}} + \mathbf{v}_\ell^{\text{love}}, \quad (3)$$

where $\mathbf{v}_\ell^{\text{self}} = c_1 r_1 (\mathbf{x}_{best} - \mathbf{x}_\ell(t))$ attracts the leader towards the global best found so far (memorized as $\mathbf{x}_{best}$). $\mathbf{v}_\ell^{\text{love}}$ is computed as in (2) from other agents, but scaled by a leader-specific love factor γ_ℓ .

2) *Global Love Attraction from Leader*: All non-leader agents experience an additional love force towards the leader, termed **global love**:

$$\mathbf{v}_i^{\text{global}} = G(t) \cdot (\mathbf{x}_\ell(t) - \mathbf{x}_i(t)), \quad (4)$$

where $G(t)$ is the global love intensity, dynamically adjusted based on swarm diversity. This term is added to the agent's velocity, so the total movement becomes:

$$\mathbf{v}_i = \mathbf{v}_i^{\text{love}} + \mathbf{v}_i^{\text{global}} + \epsilon \mathbf{r}. \quad (5)$$

3) *Adaptive Global Love and Nullification Parameters*: We define swarm diversity at iteration t as the average pairwise distance:

$$D(t) = \frac{2}{N(N-1)} \sum_{i=1}^{N-1} \sum_{j=i+1}^N \|\mathbf{x}_i - \mathbf{x}_j\|. \quad (6)$$

The global love intensity $G(t)$ adapts according to:

$$G(t) = G_{\max} \cdot \left(1 - \frac{D(t)}{D_0}\right), \quad (7)$$

where D_0 is the initial diversity. When diversity is high (exploration), $G(t)$ is low, allowing agents to explore locally. As diversity drops (convergence), $G(t)$ increases, pulling agents towards the leader to refine the solution. The nullification threshold ϕ is also adapted: $\phi(t) = \phi_0 \cdot (D(t)/D_0)^\kappa$, ensuring that when swarm is diverse, nullification is stricter (removing weak agents), and when converging, it becomes more lenient to preserve agents near the optimum.

C. Algorithm Pseudo-code

Algorithm 1 summarizes the LSNE procedure.

IV. EXPERIMENTAL SETUP AND RESULTS

A. Benchmark Functions

We evaluate LSNE on 23 classic benchmark functions [9]: unimodal (f_1 – f_7), multimodal (f_8 – f_{13}), and fixed-dimension multimodal (f_{14} – f_{23}). Functions include Sphere, Schwefel, Rosenbrock, Rastrigin, Ackley, and others. Search ranges and dimensions follow standard settings ($D = 30$ for f_1 – f_{13} , fixed for the rest). The optimization goal is minimization.

Algorithm 1 Love Swarm Nullification Enhanced (LSNE)**Require:** Population size N , parameters $\alpha, \beta, \theta, \phi_0, G_{\max}, \kappa, T_\ell, \delta, c_1$, max iterations $T_{\max}$.**Ensure:** Best solution found $\mathbf{x}_{best}$.

- 1: Initialize population $\{\mathbf{x}_i\}_{i=1}^N$ randomly in search space.
- 2: Evaluate fitness $f(\mathbf{x}_i)$ for all i .
- 3: Set $\mathbf{x}_{best} = \arg \min f(\mathbf{x}_i)$ (for minimization).
- 4: Select leader $\ell = \arg \min f(\mathbf{x}_i)$, tenure counter $t_\ell = 0$.
- 5: **for** $t = 1$ **to** $T_{\max}$ **do**
- 6: Compute diversity $D(t)$.
- 7: Update $G(t)$ and $\phi(t)$ using $D(t)$.
- 8: **Leader movement:** Compute $\mathbf{v}_\ell^{\text{self}}$ and $\mathbf{v}_\ell^{\text{love}}$; update $\mathbf{x}_\ell$ via (3).
- 9: Evaluate $f(\mathbf{x}_\ell)$; update $\mathbf{x}_{best}$ if improved.
- 10: **for** $i = 1$ **to** N , $i \neq \ell$ **do**
- 11: Compute love matrix L_{ij} for all $j \neq i$.
- 12: Identify loved agents set $\mathcal{L}_i$ (where $L_{ij} > \theta$).
- 13: Compute $\mathbf{v}_i^{\text{love}}$ (Eq. (2)) and $\mathbf{v}_i^{\text{global}} = G(t)(\mathbf{x}_\ell - \mathbf{x}_i)$.
- 14: $\mathbf{v}_i = \mathbf{v}_i^{\text{love}} + \mathbf{v}_i^{\text{global}} + \epsilon \cdot \text{rand}(-1, 1)$.
- 15: $\mathbf{x}_i \leftarrow \mathbf{x}_i + \mathbf{v}_i$; apply boundary control.
- 16: Evaluate $f(\mathbf{x}_i)$; update $\mathbf{x}_{best}$ if better.
- 17: **end for**
- 18: **Nullification:** For each agent i , compute received love $R_i = \sum_{j \neq i} L_{ji}$.
- 19: If $R_i < \phi(t)$ and $i \neq \ell$, nullify agent i : replace $\mathbf{x}_i$ with new random position (or near random loved agent).
- 20: **Leader replacement:** If $f(\mathbf{x}_{best})$ improved over current leader by $> \delta$ and $t_\ell \geq T_\ell$, set $\ell = \arg \min f$, $t_\ell = 0$; else $t_\ell \leftarrow t_\ell + 1$.
- 21: **end for**
- 22: **return** $\mathbf{x}_{best}$.

B. Compared Algorithms and Parameter Settings

We compare LSNE with:

- Original Love Swarm (LS) [4]
- Love Swarm Nullification (LSN) [5]
- Particle Swarm Optimization (PSO) [1] with inertia weight $w = 0.729$, $c_1 = c_2 = 1.494$
- Grey Wolf Optimizer (GWO) [6]
- Salp Swarm Algorithm (SSA) [7]

LSNE parameters: $N = 50$, $\alpha = 0.5$, $\beta = 0.1$, $\theta = 0.01$, $\phi_0 = 0.05$, $G_{\max} = 0.8$, $\kappa = 0.5$, $T_\ell = 5$, $\delta = 10^{-4}$, $c_1 = 1.0$, $\epsilon = 0.01$. All algorithms run for $T_{\max} = 1000$ iterations, 30 independent runs per function.

C. Results and Analysis

Table I reports mean best fitness and standard deviation for selected functions. LSNE consistently achieves the best or competitive results, especially on multimodal functions where the leader's global guidance prevents stagnation.

Convergence curves (Fig. 1) show that LSNE converges faster than LSN and LS, and often faster than PSO/GWO in early stages, while avoiding premature convergence. The

TABLE I
MEAN BEST FITNESS (STD) OVER 30 RUNS. BEST RESULTS IN BOLD.

Func.	LS	LSN	PSO	GWO	
f_1 (Sphere)	2.3e-3 (8e-4)	7.8e-5 (2e-5)	1.1e-4 (3e-5)	3.2e-6 (9e-7)	2
f_9 (Rastrigin)	27.3 (5.1)	18.9 (4.2)	45.1 (9.3)	8.7 (2.5)	
f_{10} (Ackley)	0.12 (0.03)	0.09 (0.02)	0.21 (0.06)	0.05 (0.01)	
f_{12} (Penalized)	0.005 (0.001)	0.003 (0.001)	0.042 (0.012)	0.001 (0.0004)	
f_{16} (Shekel)	-10.12 (0.05)	-10.15 (0.04)	-10.05 (0.11)	-10.16 (0.03)	

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Fig. 1. Convergence curves for selected functions (mean over 30 runs). LSNE shows rapid initial drop and sustained improvement.

leader's global attraction pulls agents out of local optima, and the adaptive $G(t)$ moderates exploitation as diversity shrinks.

D. Ablation Study: Leader Impact

We test LSNE without the leader (i.e., $G(t) = 0$ and no leader update), which reduces to an LSN variant. Table II shows significant degradation, especially on multimodal functions, confirming the leader's importance.

E. Parameter Sensitivity

We examined the leader intensity $G_{\max}$ and tenure T_ℓ . The algorithm is robust for $G_{\max} \in [0.5, 0.9]$ and $T_\ell \in [3, 10]$. Too small T_ℓ causes leader churn; too large delays adaptation to better discovered regions.

V. CONCLUSION AND FUTURE WORK

We introduced LSNE, a novel swarm optimizer that enhances the Love Swarm Nullification algorithm by incorporating a leader agent with global love attraction. The leader provides directional guidance, while emotional interactions among agents maintain diversity. Adaptive mechanisms for love intensity and nullification further balance exploration and exploitation. Experiments on 23 benchmarks demonstrate

TABLE II
ABLATION: LSNE WITH AND WITHOUT LEADER (MEAN FITNESS).

Function	LSNE (full)	LSNE w/o leader
f_9 (Rastrigin)	5.2	21.4
f_{10} (Ackley)	2.3e-5	0.09
f_{13} (Penalized2)	8.1e-6	0.0025

LSNE’s superior convergence speed and accuracy compared to original Love Swarm variants and state-of-the-art methods. Ablation studies confirm the leader’s essential role.

Future work includes extending the leader concept to multiple hierarchical leaders (like a “love council”), applying LSNE to constrained and multi-objective optimization, and investigating the algorithm’s swarm dynamics using network theory.

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